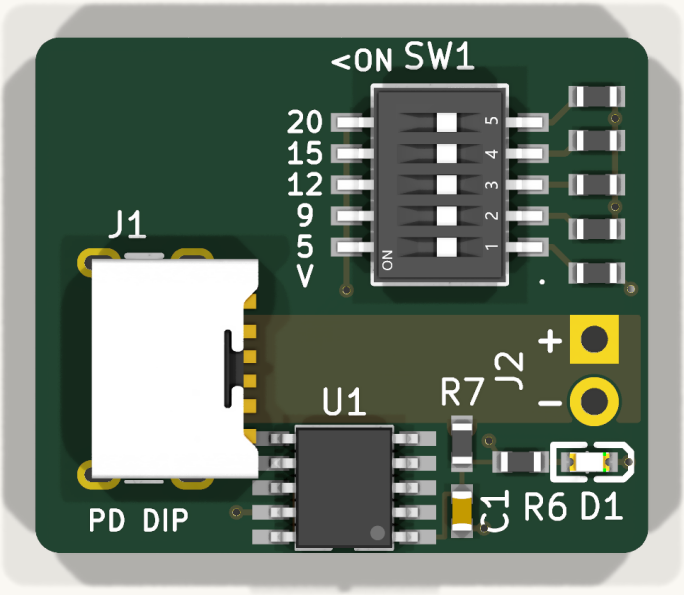
pd-trigger-lite-dip

A 25 by 21 mm USB-C PD trigger: flip one DIP switch to get 5, 9, 12, 15 or 20 V at up to 3 A on two solder pads.

This board asks a USB-C Power Delivery charger for one fixed voltage and passes the charger's VBUS straight to two 2.54 mm output holes. There is no regulator in the power path, so the output is exactly what the charger delivers. You pick the voltage with a five-position DIP switch, one switch per voltage, so changing it needs no soldering iron. It is the pd-trigger-lite board with its solder links replaced by that switch, and 5 V added as a choice.



TOP VIEW 25 X 21 MM, 2 LAYERS, 1 OZ RENDER

USB-C receptacle on the left, the voltage switch SW1 at the top with the voltage printed beside each position, the PD controller at the bottom, output pads on the right (+ is VBUS, - is ground). The red LED sits under the output pads.

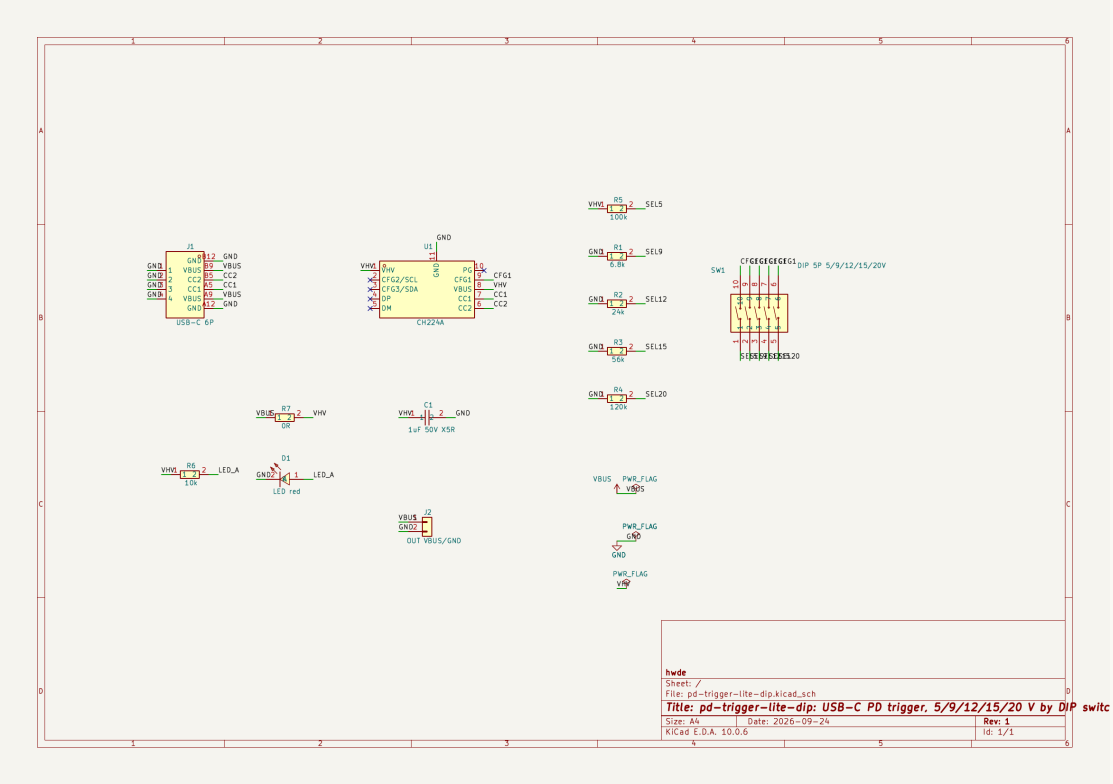
© 2026 IHSAN SALARI. MIT LICENSE.

Schematic

One controller, one capacitor, one switch and five resistors. Everything else is copper.

The receptacle J1 is a 6-pin power-only part: two VBUS contacts, two ground contacts and the two CC lines. The controller U1 (WCH CH224A) talks to the charger over CC1 and CC2. It is powered from VBUS through the 0 ohm link R7, which only exists so the controller's pins can be reached with thin copper, and C1 (1 uF, 50 V) decouples it. VBUS runs as a wide copper pour from J1 to the output pad J2.

U1 reads its CFG1 pin at power-up. One side of every switch in SW1 is CFG1; the other side goes to that position's resistor. Position 1 pulls CFG1 up to VHV through R5 (100k), which the CH224 manual lists as the way to request 5 V. Positions 2 to 5 tie CFG1 to ground through R1 to R4 (6.8k, 24k, 56k, 120k), the manual's single-resistor settings for 9, 12, 15 and 20 V. CFG2 and CFG3 are left open; the chip pulls them up inside. R6 and D1 light the LED from VBUS.



SCHEMATIC KICAD/PD-TRIGGER-LITE-DIP.KICAD_SCH ERC 0 ERRORS, 0 WARNINGS

Nets join by label name. SEL5 to SEL20 are the nodes between each resistor and its switch.

© 2026 IHSAN SALARI. MIT LICENSE.

Setting the voltage

Exactly one switch ON, set before you plug the charger in.

SWITCH ON	OUTPUT	OTHERS	WHAT IT DOES
1	5 V	all OFF	R5 100k pulls CFG1 up to VHV
2	9 V	all OFF	R1 6.8k from CFG1 to ground
3	12 V	all OFF	R2 24k from CFG1 to ground
4	15 V	all OFF	R3 56k from CFG1 to ground
5	20 V	all OFF	R4 120k from CFG1 to ground
none	undefined		CFG1 floats; do not use
two or more	not allowed		can damage U1 (see below)

On the board the voltage is printed beside each switch: 20, 15, 12, 9 and 5 from the top, so position 1 (5 V) is the bottom one, nearest the output pads. A switch is ON when its slider is pushed toward the side marked ON on the switch body, the edge opposite its printed 1 to 5. The board's silkscreen marks the same side "<ON" (toward the voltage labels). Use a toothpick or a small screwdriver; the sliders are 1.27 mm apart.

EXACTLY ONE SWITCH, SET BEFORE PLUGGING IN

The controller reads the switch once, when it powers up, so set it with the board unplugged and re-plug after any change. With every switch OFF, CFG1 floats and the request is undefined. Never turn on two at once: with switch 1 and any other ON, R5 and the ground resistor form a divider from VHV, and once VHV rises above 5 V that can push CFG1 past its 3.8 V absolute maximum.

Output. The two plated holes take 20 AWG wire or a 1x2 2.54 mm header, which you solder yourself; JLC does not fit them. The rating is 3 A at any voltage. A charger without an e-marked cable never offers more than 3 A, and the board makes no 5 A claim.

LED. D1 is lit whenever VBUS is present. Its current rises with the voltage, so it glows faintly at 5 V and clearly at 12 V and above.

First power-up. Plug in with nothing on the output, measure the voltage across the pads, and only then connect the load. The charger must offer the chosen voltage as a fixed PDO; many phone chargers stop at 9 V, and a charger that lacks it leaves the output at 5 V. The board has no fuse and no TVS diode, because the controller's datasheet asks for neither. A

© 2026 IHSAN SALARI. MIT LICENSE.

hot-plug spike at 20 V is the one plausible way to damage it, so at 20 V plug the charger into the wall first and the cable into the board last.

© 2026 IHSAN SALARI. MIT LICENSE.

What is on it

Twelve parts placed by JLC, three of them Extended. Everything else is bare copper.

REF	PART	LCSC	TYPE	QTY
U1	CH224A PD sink controller, ESSOP-10	C42459160	Extended	1
J1	USB-C receptacle 6P, 3 A 20 V (TYPE-C-6M-001)	C2798175	Extended	1
SW1	DIP switch, 5 positions, 1.27 mm SMD (SHOU HAN 1.27-5P TPPT)	C7421518	Extended	1
C1	1 uF 50 V X5R 0603	C15849	Basic	1
R1	6.8k 1% 0603 (9 V)	C23212	Basic	1
R2	24k 1% 0603 (12 V)	C23352	Basic	1
R3	56k 1% 0603 (15 V)	C23206	Basic	1
R4	120k 1% 0603 (20 V)	C25808	Basic	1
R5	100k 1% 0603 (5 V)	C25803	Basic	1
R6	10k 1% 0603	C25804	Basic	1
R7	0 ohm 0603	C21189	Basic	1
D1	red LED 0603	C2286	Basic	1

Not assembled: J2, the pair of output holes. The three Extended parts each add JLC's per-part feeder fee; no Basic part can replace any of them.

What it costs

About 33 USD for five assembled boards before shipping, as an estimate.

27.42

USD, JLC estimate for 5 boards: PCB 2.00, setup 8.00, 3 Extended fees 9.00, stencil 8.00, joints 0.42.

5.44

USD, parts for 5 boards at LCSC unit prices, about 1.09 per board (the switch is 0.53).

32.9

USD total for 5 boards, before shipping and tax.

© 2026 IHSAN SALARI. MIT LICENSE.

These are estimates from the skill's price table (verified 2026-07-24), not a quote. Live JLC prices have run 1.9 to 3.1 times higher than that table for bare PCBs, so expect the PCB line to be nearer 4 to 6 USD. Shipping depends on the country and the carrier and is not included. The cart at `https://cart.jlcpcb.com/quote` is the only real price.

Estimate from fab/quote.json (order_quote.py, qty 5, HASL, economic assembly) and kicad/parts.json prices, read 2026-09-24.

© 2026 IHSAN SALARI. MIT LICENSE.

Ordering at JLCPCB

Three files go up: the gerber zip, then BOM.csv and CPL.csv for assembly.

Pre-buy list

JLC's BOM review may show an Extended part as idle stock: unselected, at quantity 0, until it has been bought into your JLC parts inventory. JLC's public parts search cannot tell idle stock from stock that assembles straight away, so all three Extended parts are listed. For 5 boards:

LCSC	PART	REF	QTY TO BUY
C2798175	USB-C receptacle 6P (TYPE-C-6M-001)	J1	5
C7421518	DIP switch 5P, 1.27 mm (1.27-5P TPPT)	SW1	5
C42459160	CH224A PD sink controller	U1	5

From fab/prebuy.csv (bom_cpl.py, build of 5 boards).

1. Open the JLC quote page (link above) and upload the gerber zip, `fab/pd-trigger-lite-dip_gerbers.zip`.
2. Check that the viewer reads 2 layers, 25 x 21 mm. Keep 1.6 mm, 1 oz, HASL, green, quantity 5.
3. Turn on PCB Assembly: Economic, top side, 5 boards.
4. Upload `fab/BOM.csv` and `fab/CPL.csv`. Confirm all twelve parts matched their LCSC numbers and that J2 is absent, since it is not assembled.
5. If a part on the pre-buy list shows as unselected at quantity 0, buy it into your parts inventory from the BOM review page, then select it and re-check the quantity.
6. In the placement preview, check four parts. SW1: its pin-1 mark sits at the bottom-right corner, next to R5, so switch 1 is the 5 V row and the switch's ON edge faces the voltage labels. D1: its cathode mark must face the right board edge. U1: the pin-1 dot sits at the corner nearest the capacitor C1. J1: the receptacle mouth points off the left edge. No rotation correction was applied to the CPL, because every footprint comes from JLC's own library.
7. Run JLCDFM on the gerbers as a second opinion, then pay. Payment is yours to do.

© 2026 IHSAN SALARI. MIT LICENSE.

Gates on 2026-09-24: ERC 0/0, DRC 0 errors 0 unconnected, place, verify and DFM pass.

What it cost to design

About 6 USD of model spend went into designing this board. It is not part of the board's price.

STEP	USD
Before the workspace (brief and skill read)	1.13
Schematic	0.85
Placement	2.10
DFM, order files and guide	2.23
Total	**6.31**

The split follows when hwde recorded each step, and the rows are rounded to the cent. A later round that wrote the pre-buy list and fixed this board's BOM cost 4.93 USD more, including an iteration that timed out. It is not in the total because it also built the pre-buy list feature for every board. The review-fix round resumed the design session, so its cost is already inside the total. Later skill-wide rebuilds of this guide are skill work and are not counted.

reports/cost.json (gen_cost.py), priced from reference/model_prices.yaml, which was fitted to Claude Code's own session totals on 2026-09-24.

© 2026 IHSAN SALARI. MIT LICENSE.